

Microwave electrometry with Rydberg atoms in a vapour cell using bright atomic resonances

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Atom-based standards for length and time as well as other physical quantities such as magnetic fields show clear advantages by enabling stable and uniform measurements. Here we demonstrate a new method for measuring microwave (MW) electric fields based on quantum interference in a rubidium atom. Using a bright resonance prepared within an electromagnetically induced transparency window we could achieve a sensitivity of $\sim 30 \mu\text{V cm}^{-1} \text{ Hz}^{-1/2}$ and demonstrate detection of MW electric fields as small as $\sim 8 \mu\text{V cm}^{-1}$ with a modest set-up. The sensitivity is limited, at present, by the stability of our lasers and can be significantly improved in the future. Our method can serve as a new atom-based traceable standard for MW electrometry, with its reproducibility, accuracy and stability promising advances towards levels comparable with those attained in magnetometry at present.

The reproducibility, accuracy and stability of quantum objects, such as atoms, has made many of the standards and measurement techniques established early in the past century relics. Most progress in adopting atoms for standards has been made in measuring atomic transition frequencies that are now used for length and time standards. Atomic clocks have now realized an accuracy of better than one part in 10^{14} (ref. 1). Concurrently, tremendous progress has also been made in measuring magnetic fields, where it is now possible to achieve an accuracy of $\text{fT Hz}^{-1/2}$ (refs 2–5). Motivated by applications of these advances, such as g-factor measurements⁶ and the global positioning system, it is straightforward to conclude that quantum-system-based sensing is relevant to pursue.

In contrast to the successes already mentioned, comparatively little progress has been made in applying atoms to measure electric fields, particularly using methods that are practical to take into the field outside a laboratory, despite the clear advantages and the ubiquity of electric fields. The basic concepts used to measure a MW electric field and the traceable standards for calibrating those devices have changed little from the ones Hertz pioneered in the 1880s (ref. 7). The present traceable standards for MW electric field measurement are referred to as the standard antenna and standard field methods^{8,9}. For fields up to 40 GHz, these techniques use a resistively loaded dipole antenna and a MW diode detector. Electric fields at a level of $\sim 1 \text{ mV cm}^{-1}$ can be determined and used for calibration^{10–13}. Modern variations on sensing MW electric fields for traceable standards are based on optical measurements of the electro-magnetic fields converted by an antenna. These set-ups can sense MW electric field strengths down to $\sim 30 \mu\text{V cm}^{-1}$ (refs 14,15) with a sensitivity of $\sim 1 \text{ mV cm}^{-1} \text{ Hz}^{-1/2}$ (ref. 10). In all of these types of measurement, the voltage induced in the probe has to be used to back calculate the electric field. For MW electric field sensing, then, a major limitation is the antenna, as it is the converter of the MW electric field to some observable, for example voltage, and depends on geometry, can lead to perturbations of the MW electric field, particularly in the near field, and can suffer from

out-of-band interference. The optical detection system is subject to ageing and manufacturing variations.

The accurate measurement of electric fields is important for a host of applications, especially at MW frequencies. Perhaps most importantly, the ability to sensitively measure MW electric fields can allow for the amplitude stabilization of a MW electric field source and the determination of optical properties of materials at these frequencies to high precision. Previous work in this direction has been termed the development of an atomic candle^{16,17}. The K_u -band MWs that we use in our experiments, and K-band MWs generally, are particularly useful for experimental high-resolution weather radar and satellite communications. A portable traceable standard is important for field calibrating systems such as these¹⁸.

We demonstrate a method for making a traceable MW electric field sensor based on ^{87}Rb Rydberg atoms excited in a glass vapour cell. The working principle of the sensor is based on detecting how MW electric fields affect the optical transitions of ^{87}Rb Rydberg atoms. Using electromagnetically induced transparency (EIT), the MW electric field is detected as a bright resonance within the EIT signal. The minimum detected MW electric field and sensitivity are already superior to present traceable methods. Optical readout allows for spatial resolution in the micrometre regime and facilitates near-field measurements. The basic concept can be scaled up to larger one- or two-dimensional arrays that include fibre optical delivery of the light for readout. The set-up and detection scheme is conducive for miniaturization and portability, particularly using microcells¹⁹. As such, it can become a practical traceable MW electric field sensor. The technique presented here can also yield information about the polarization of the field but the details of this aspect of the method lie outside the scope of this work.

Approach

EIT is a quantum interference process in which two excitation pathways of a three-level system interfere to produce transmission on an atomic resonance^{20,21}. The quantum interference produces a dark state that does not absorb light under normal resonance

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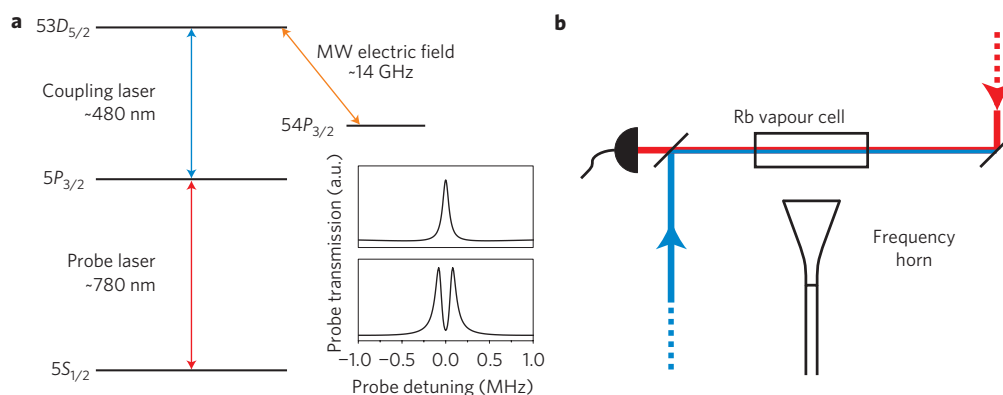


Figure 1 | Level diagram and experimental set-up. a, The energy level diagram for the four-level system used for the experiments. The top part of the inset shows an example EIT feature associated the three-level system without a MW electric field. The bottom part of the inset shows an example of the bright resonance that is produced within the EIT window when a MW electric field is present. **b**, The experimental set-up used for the experiments.

conditions. As EIT depends on quantum interference, it is exquisitely sensitive to phase disturbances, transitions out of the participating states and energy level shifts of the three-level system. The addition of a MW electric field that is resonant with a nearby Rydberg transition, a fourth level, for the scheme shown in Fig. 1a, breaks the symmetry of the EIT interference and can produce a spectrally sharp bright resonance within the EIT line shape. In a similar configuration, MW electric fields have been used to dress Rydberg atom transitions to enhance their sensitivity to d.c. electric fields and manipulate Rydberg atom interactions^{22,23}. The ability to detect a line-shape change of the EIT transmission window is fundamentally limited by the laser linewidths, transit-time broadening, shot noise and the decay and dephasing rates of the Rydberg states involved in the EIT process. The strength of the effect depends on the coupling to the MW electric field, which is determined by the Rabi frequency of the transition, $\Omega_{\text{MW}} = \mu_{\text{MW}} E / \hbar$, where E is the amplitude of the MW electric field and μ_{MW} is the transition dipole moment. Although technically challenging, it is possible, in principle, to make MW electric field measurements of $\leq 100 \text{ nV cm}^{-1}$.

Rydberg atoms are atoms in highly excited states with large principal quantum numbers n , and long lifetimes²⁴. The associated large polarizability of Rydberg atoms has been used by several groups to measure d.c. electric fields at the level of $\geq 100 \mu\text{V cm}^{-1}$ using field ionization and atomic beams^{25,26} and EIT with ultracold atomic gases²⁷. A minimum d.c. electric field measurement of $\sim \pm 20 \mu\text{V cm}^{-1}$ has been achieved^{28,29}. Fluorescence imaging has yielded d.c. electric field measurements around 10 V cm^{-1} (ref. 30). In contrast, our MW electric field measurements rely on resonant transitions and the associated large transition dipole moment between neighbouring Rydberg states that scales as n^2 , $\mu_{\text{MW}} \sim 4000 ea_0$ (where e is the electron charge and a_0 the Bohr radius) for $n \sim 65$ in ^{87}Rb . The sensitivity to MW electric fields can be resonantly enhanced for frequencies in the range of 1–500 GHz, as there are many resonant electric dipole transitions between Rydberg states in this frequency band that can be Stark shifted to different frequencies with d.c. electric fields. As a consequence of the large transition dipole moment, the MW electric field coupling of two close-lying Rydberg states, Ω_{MW} , can be large when the MW electric field is weak. As an example, consider the $^{87}\text{Rb } 55D \rightarrow 54F$ transition at a frequency of $\sim 13.9 \text{ GHz}$. A beam intensity of 5 fW cm^{-2} corresponding to a MW electric field amplitude of $64 \mu\text{V cm}^{-1}$ yields $\Omega_{\text{MW}} \sim 1 \text{ MHz}$. A small MW electric field amplitude results in a signal that is straightforward to observe with present frequency-stabilized diode lasers and room-temperature vapour cells. Our all-optical, vapour cell method can achieve similar sensitivities and is technically much easier to realize than Rydberg atom masers

and atomic beam methods used in the millimetre-wave regime. These sensors require ultrahigh vacuum and even coupling of the radiation into a high-Q millimetre-wave cavity in the apparatus^{31,32}.

As each atom is identical in structure, our approach uses each non-interacting, independent atom that participates as an identical, stable atomic sensor of the MW electric field. The device is traceable because its sensitivity is directly linked to the properties of the atom, namely the atomic Rydberg wavefunctions or dipole moments that are well known and can be determined even more precisely with modern spectroscopy, for example using frequency combs and ultracold atoms to better determine the Rydberg atom quantum defects or using Stark shift measurements. The properties of Rydberg atoms are readily calculated to high precision. Dipole moments can be determined to a level of $10^{-4} ea_0$ at present³³, suggesting that the Rydberg atomic dipole moments can be determined to one part in 10^8 using the present methods.

Discussion

Measured probe laser transmission curves as a function of probe laser detuning for the four-level system found in Fig. 1a are shown in Fig. 2. Traces for several different MW electric field amplitudes are shown. At small MW electric field amplitudes (Fig. 2a) the MW electric fields cause a broadening and reduction of the peak EIT transmission signal. The linewidth of the EIT transmission window is $\sim 4\text{--}5 \text{ MHz}$, whereas the relative drift of the coupling and probe lasers is $\leq 2 \text{ MHz h}^{-1}$ determined from measurements in the absence of a MW electric field. At intermediate MW electric field amplitudes, an absorption dip corresponding to the bright resonance appears within the EIT transmission window. The frequency for the transition is calculated to be 14.2318 GHz using the quantum defects from refs 34,35, in agreement to 0.008% with the experimentally observed frequency of $14.2330 \pm 0.0001 \text{ GHz}$. At large MW electric field amplitudes (Fig. 2b) the EIT transmission peak is split by the MW electric field. In this case, the peak splitting can be interpreted as Autler–Townes splitting of the $53D_{5/2}$ state and is relatively large because the calculated radial transition dipole moment is $\mu_{\text{MW}} = 3611 ea_0$. For comparison, μ_{MW} is more than three orders of magnitude larger than the transition dipole moments on the coupling and probe transitions.

Without Doppler averaging, the splitting of the peaks in Fig. 2b is equal to the Autler–Townes splitting of the states, $\Omega_{\text{MW}}/2\pi$. Doppler averaging changes the splitting of the peaks by the ratio of the coupling laser to probe laser wavelength when the probe laser is scanned and the coupling laser is locked on the unshifted resonance. The effect results from the fact that the $53D_{5/2}$ state is split by Ω_{MW} , which detunes the coupling laser from resonance. The probe laser resonance condition is then shifted by the ratio of

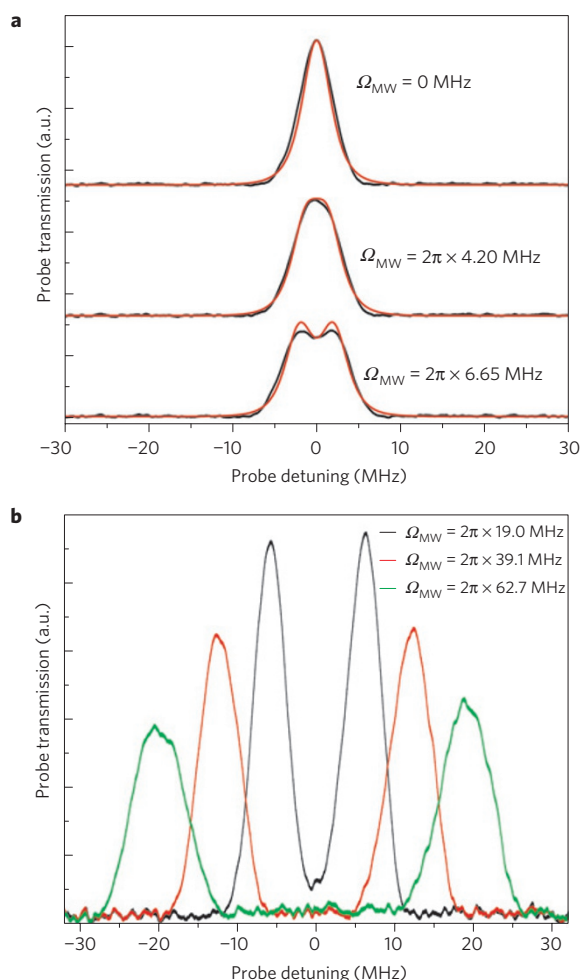


Figure 2 | Three-level EIT and splitting from the MW electric fields. **a**, The experimental bright resonance dip and attenuation of the EIT transmission signal for low MW electric field amplitudes (black) with theory curves (red). The differences in the theory and experiment in the lower curves are attributed to the fact that we are using a four-level theory in these plots rather than the full 52-level theory and to uncertainties in the transit-time broadening. **b**, Autler-Townes splitting of the $53D_{5/2} \rightarrow 54P_{3/2}$ Rydberg transition that occurs for larger MW electric field strengths. The experimental parameters can be found in Methods. The peak splitting corresponds to $\lambda_c/\lambda_p \Omega_{MW}/2\pi$. The uncertainties in Ω_{MW} are described in the text.

the wavelengths. In our case, with $\lambda_p \simeq 780$ nm and $\lambda_c \simeq 480$ nm, the Doppler averaged peaks are separated by $\lambda_c/\lambda_p \times \Omega_{MW}/2\pi = 0.615 \times \Omega_{MW}/2\pi$. In a vapour cell, the ratio of these wavelengths needs to be known accurately to make precise measurements of the MW electric field. The D_2 transition wavelength is known to one part in 10^{11} (ref. 36) and wavelengths can be determined to one part in 10^8 with commercial laboratory wave meters, so this effect is not limiting at this point. The effect can, in principle, be used to amplify the observed Autler-Townes splitting if, for example, a set of transitions with a probe wavelength longer than the coupling wavelength is used or the coupling laser is scanned.

Using the Autler-Townes splitting and the calculated transition dipole moment, the MW electric field inside the cell is known with an uncertainty of 0.5%, determined, in our case, by the precision with which the splitting can be measured at present. The MW electric field strengths obtained in this way are in agreement within 10% of those calculated using the parameters of the resonant horn antenna, including its position relative to the Rb vapour cell. We

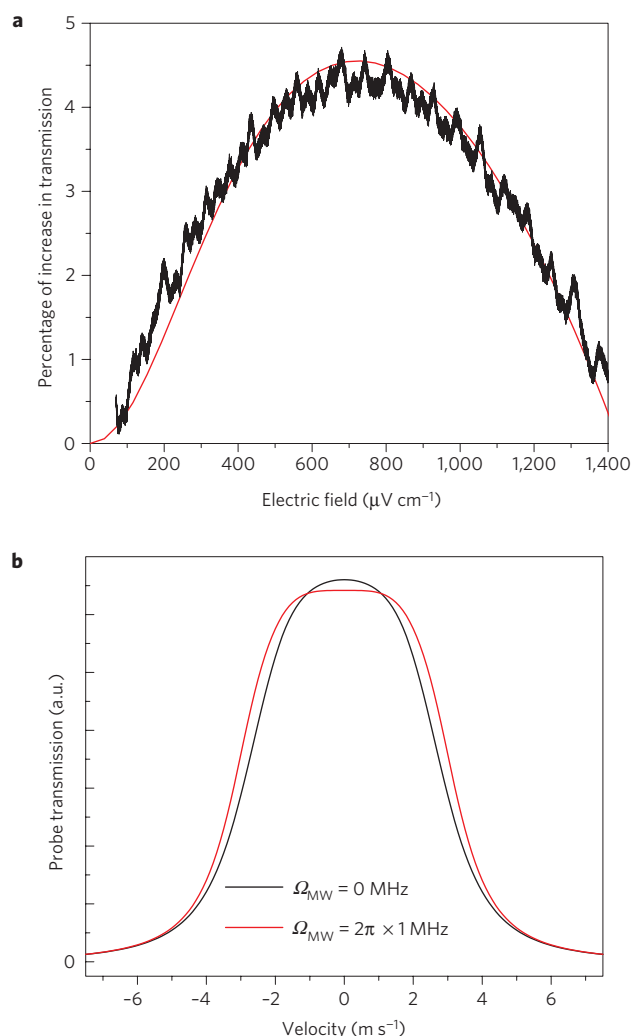


Figure 3 | Enhanced transmission of the four-level EIT signal due to the MW electric field. **a**, Probe transmission on resonance due to the MW electric field. The experimental results show agreement with a four-level density matrix calculation. The red line is the theory and the black line is the experimental data. The parameters for the measurements are provided in Methods. The transmission at the line centre will continue to decrease as the electric field increases to the Autler-Townes regime. The MW electric field is tuned 1 MHz off resonance. **b**, The enhancement of the EIT transmission when the MW electric field is applied for small MW electric field amplitudes. The MW electric field broadens the two-photon EIT transition in velocity space in a thermal vapour. This leads to enhanced transmission of the probe laser.

believe the difference between the calculated and measured MW electric fields is mainly due to the uncertainty associated with the characterization of the horn antenna and the MW generator. The disagreement is constant to within 0.55% over the range of MW electric fields studied. In addition, at low MW electric fields, there is an uncertainty in setting the output power of the MW generator of 4.4% (ref. 37).

Density matrix calculations for the four-level system, described in Methods, are also shown in Fig. 2. The agreement between the four-level theoretical calculations and the data shows that the system is modelled well. Our full 52-level density matrix calculations that include all of the hyperfine states, $|F m_F\rangle$, of each of the participating levels show that the populations of states $53D_{5/2}$ and $54P_{3/2}$ get optically pumped into the stretched $|4, 4\rangle$ and $|3, 3\rangle$ states, respectively, consistent with other works³⁸. Our

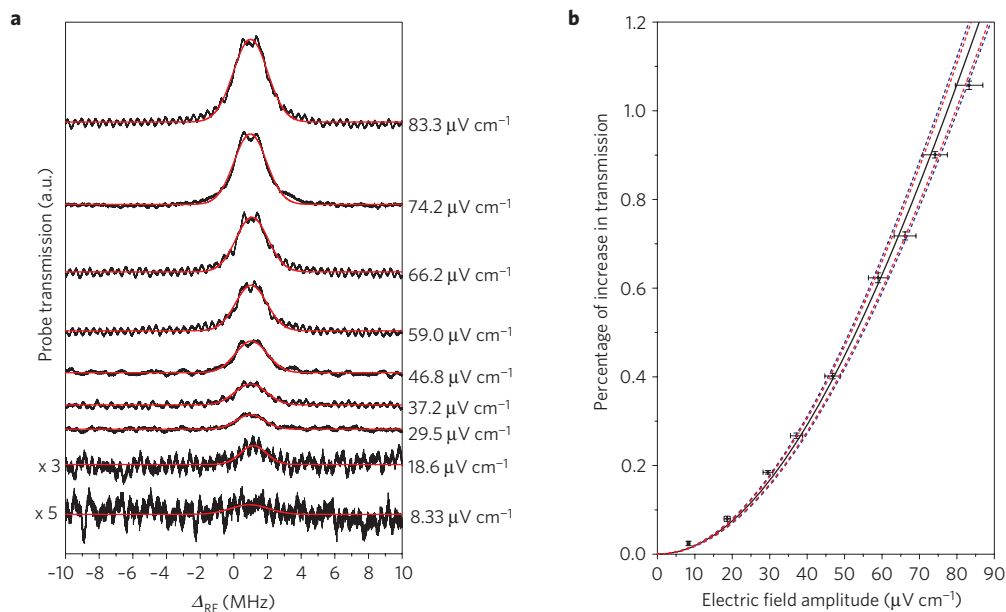


Figure 4 | Four-level EIT transmission signal as a function of MW electric field amplitude. **a**, Transmission line shapes for different values of MW electric field amplitude as a function of MW electric field detuning (Δ_{RF}). The black curves are experimental data, and the red curves are Gaussian fits to the data. The graph shows the change in probe transmission as a function of MW electric field frequency. All of the data shown were taken under the same conditions as in Fig. 3, except the MW electric field was tuned to resonance. Each trace is the average of 9,000 scans. **b**, The on-resonance peak height taken from the traces in **a** as a function of MW electric field amplitude. The dashed red line is a theoretical four-level density matrix calculation. The vertical error bars in the figure represent the standard deviation of the fits to the peak heights. The horizontal axis is derived from the MW power applied to the horn antenna and the associated experimental geometry. The horizontal error bars are the uncertainty of calculating the MW electric field due to uncertainty in the output field of the MW generator, 5% accounting for both the calibration error to the Autler-Townes splitting at larger MW electric fields and the output setting error described in the text. The dashed blue (red) lines represent theoretical variation due to a change of $\pm 10\%$ in the coupling (probe) power.

four-level model uses these states for calculating the angular part of the transition dipole moment. The deviation of the theory from the experiment is a result of using a four-level theory to describe a multilevel system with different branching ratios to the transitions and the uncertainty in the beam shape, which influences the transit-time broadening. The attenuation and splitting of the EIT transmission peak shown in Fig. 2 is the signal we propose to use to precisely measure MW electric fields.

Figure 3a shows the peak height of the EIT transmission window as a function of MW electric field strength. In this experiment the lasers are locked to the EIT peak and the MW electric field strength is swept in time while the transmission of the probe beam is monitored. The red line shows the four-level density matrix calculation for this range of MW electric field strengths. For these MW electric fields, the probe transmission increases from the three-level EIT value. A maximum increase of $\sim 4.5\%$ is shown in Fig. 3a. For our experimental parameters and MW electric field amplitudes less than $\sim 1,400 \mu\text{V cm}^{-1}$, the probe transmission is increased in the presence of the MW electric fields with a maximum at $\sim 715 \mu\text{V cm}^{-1}$. The increase in probe transmission is also a result of the Doppler effect. When the MW electric field is small, the coupling to $54P_{3/2}$ is small and there is little population in this state. However, because the MW electric field wavelength is comparatively long and therefore less sensitive to Doppler shifts, the dominant effect of the MW electric field is to shift, on average, the atoms participating in the EIT to smaller two-photon detunings. This increases the EIT transmission signal on resonance. Figure 3b shows a plot of the calculated probe transmission as a function of velocity for $\Omega_{MW} = 0$ MHz and $\Omega_{MW} = 2\pi \times 1$ MHz to illustrate this effect. Although the probe transmission is suppressed for atoms near zero velocity, the suppression is more than made up for by the increased participation of atoms in the wings of the line.

The position of the peak transmission in Fig. 3a can be shifted by changing either the probe or coupling Rabi frequency.

Figure 4 shows the central result of this paper. In Fig. 4a, a series of traces of the change in probe laser transmission as a function of MW frequency relative to the three-level EIT signal is presented for various MW electric field amplitudes calculated from the MW generator and geometry. The probe and coupling lasers are locked on the EIT transmission resonance. The increased transmission is due to the resonant MW electric field and not an off-resonant a.c. Stark shift. A change in the EIT transmission is observed only when the MW electric field is near resonance. The EIT transmission does not change when the MW electric field is tuned off resonance. Off-resonant or multi-photon absorption can be an issue at larger MW electric field strengths if they make a significant contribution to the probe absorption, but depend on the Rydberg states being used. When the MW power is held constant and the MW electric field frequency is scanned over the $53D_{5/2} \rightarrow 54P_{3/2}$ resonance, the maximum signal occurs at the experimental transition frequency, 14.2330 ± 0.0001 GHz. The smallest detected MW electric field shown in Fig. 4 is $8.33 \pm 0.37 \mu\text{V cm}^{-1}$, limited by the frequency and intensity stability of the probe and coupling lasers. Figure 4b shows the probe transmission peak heights as a function of MW electric field amplitude compiled for the data shown in Fig. 4a. The solid red curve shows our four-level density matrix calculation to be in excellent agreement with the data. For MW electric field amplitudes $\lesssim 100 \mu\text{V cm}^{-1}$, the maximum probe transmission signal decays smoothly, as seen in Fig. 4b. The purpose of Fig. 4b is to demonstrate that the data agree with the theory to within the statistical error of our knowledge of the test MW electric field and experimental error.

To obtain an estimate of the sensitivity of our present apparatus, we measured the probe transmission on the EIT resonance and

Table 1 | Principal error sources of the present measurements.

Effect	Description	Autler–Townes splitting	Error type	Peak amplitude	Error type
Constant magnetic field (~ 0.5 G)	Pollution of transition dipole ($\sim \mu_B B / \Delta E_{fs}$)	1%	Systematic	1%	Systematic
Constant electric field (~ 1 mV cm $^{-1}$)	Pollution of transition dipole ($\sim \alpha E^2 / \Delta E_{fs}$)	0.001%	Systematic	0.001%	Systematic
Vapour pressure/density changes ($\Delta T \sim 0.01$ °C)	Absorption changes (ref. 37)	NA (changes absorption but does not change splitting)	–	0.01%	Systematic
Microwave source frequency	Uncertainty in MW source frequency (ref. 34)	0.55%	Systematic	NA (the MW frequency was tuned to the EIT peak)	–
Microwave source amplitude	Uncertainty in amplitude at low MW fields (ref. 34)	NA (this was calibrated to the Autler–Townes splitting)	–	4.4%	Systematic
Two-photon detuning noise (200 kHz)	Relative detuning noise between coupling and probe lasers	NA (broadens spectral lines but does not change splitting at this level of noise)	–	0.4%	Statistical
Technical noise from laser intensity noise and detection	Statistical noise in fitting the peak separation or height	0.5%	Statistical	~ 1 –7% for peaks in Fig. 4, growing from largest to smallest signal	Statistical

This table lists the main sources of systematic error and statistical noise present in the Autler–Townes and peak amplitude measurements. The principal source of statistical noise in both experiments is technical noise associated with the laser frequency and amplitude stability. The principle source of systematic error is the change in dipole moment associated with the Earth's magnetic field and uncertainties associated with the frequency generator. Statistical fluctuations of the background electric and magnetic fields are negligible. The systematic error due to the Earth's magnetic field can be addressed by zeroing it with an external field or by shielding the cell so that the MW field can still be coupled into it. μ_B is the Bohr magneton, ΔT is the change in temperature over a measurement, α is the polarizability of the Rydberg state and ΔE_{fs} is the fine-structure splitting of the Rydberg state. NA, not applicable.

compared the signal in the cases where the MW electric field was on and then off. The coupling laser was modulated at 22 kHz while the MW electric field was modulated at 700 Hz. The signal was processed in two different lock-in amplifiers with an overall detection system integration time of 1 s. For a signal-to-noise ratio of ~ 1 , we were able to measure a MW electric field amplitude of 20 ± 1 μ V cm $^{-1}$. The measurement corresponds to a sensitivity of $\sim 30 \pm 2$ μ V cm $^{-1}$ Hz $^{-1/2}$.

Several sources of systematic error and statistical noise associated with our present experiments are shown in Table 1. The primary source of systematic error is due to the background magnetic field of the Earth. Future experiments will have to address this error by either zeroing the magnetic field or shielding the atoms. The dominant source of statistical error at both small and large MW electric fields is the technical noise arising from the frequency and intensity instability of the coupling and probe lasers. These fluctuations determine both the sensitivity and the maximum effective integration time. The detection is not shot-noise limited at present and can be significantly improved in the future, first, by reducing the laser technical noise.

Conclusion

We have demonstrated a new method for traceable MW electrometry based on Rydberg atom EIT. The smallest detectable MW electric field amplitude and sensitivity demonstrated are below the present antenna-based standards used for traceable MW electric field measurements by more than an order of magnitude. In addition, our method minimally perturbs the MW electric field, is a direct measure of the MW electric field linked to the Rydberg atom transition dipole moments, and is resonantly enhanced to give some immunity to out-of-band interference. The present apparatus can be improved in many ways, including the use of lasers with a narrower spectral bandwidth and better amplitude stability, the use of frequency-modulated spectroscopy, the implementation of lower noise photo-detectors, and the use of the dispersive nature of the sharp resonances used in our experiments. In addition, we

speculate that if the technique is pushed to the quantum limit it may be possible to improve it by squeezing. These improvements may push the limits of this technique to $\lesssim 100$ nV cm $^{-1}$, at least for some MW electric field frequencies. Now that a practical atomic approach has been applied to MW electrometry, the outlook is promising for adopting such a method for future traceable MW electric field standards.

Methods

The four-level energy level diagram for the atomic system is shown in Fig. 1a. The probe and coupling laser beams counter-propagate through a room-temperature 10-cm-long Rb vapour cell. The vapour pressure is $\sim 10^{-6}$ torr, yielding a total ground-state atom density, $\rho_g \sim 3.5 \times 10^{10}$ cm $^{-3}$, that is constant to within $\pm 0.05\%$. The estimated temperature drifts during the measurements are < 0.01 °C. The vapour cell is not magnetically shielded. The Earth's magnetic field, ~ 0.5 G, shifts the transition between the Rydberg states ~ 1 MHz. The MW field was tuned to the Rydberg resonance, so the main affect of the magnetic field was to change the dipole moment as the states are hybridized by the field. We observed no d.c. Stark shifts in our experiments and estimate the electric field to be ≤ 1 mV cm $^{-1}$ given the vapour cell geometry and our previous work¹⁹. This field gives ≤ 1 kHz of shift at our quantum numbers. The hybridization of the states due to the electric field is smaller than that from the magnetic field (Table 1). Stray d.c. electric fields can be minimized by heating the vapour cell. Background black-body radiation is expected to yield a.c. Stark shifts of ~ 2 kHz (ref. 24). To estimate non-resonant two-photon excitation by the MW electric field, we assumed a coincidental detuning of 10^{-1} n^{-3} (in atomic units) for each step of the two-photon transition. Using this estimate, it would take an electric field of $E = 500$ mV cm $^{-1}$ to yield a 1% excitation rate out of the Rydberg state with the effective two-photon Rabi frequency scaling as E^2 . The Rydberg atom density is kept low, $\rho_R \sim 10^7$ cm $^{-3}$, so collisions do not play a significant role. Rydberg atom interactions³⁹ yield < 100 Hz interaction for ρ_R and collisions involving Rydberg atoms and ground state atoms give ~ 2 Hz shifts in our case²⁴. The average Rabi frequency on the probe (coupling) transition is $\Omega_p = 2\pi \times 6.00 \pm 0.05$ MHz ($\Omega_c = 2\pi \times 2.00 \pm 0.04$ MHz). The probe (coupling) beam size is 750 ± 4 μ m (100 ± 4 μ m). The Gaussian focusing of the coupling laser gives an effective interaction length of 7.5 cm. The probe laser, ~ 780 nm, is locked to the $5S_{1/2}(F=2) \rightarrow 5P_{3/2}(F=3)$ 87 Rb transition. The coupling laser, ~ 480 nm, is locked to a Fabry–Perot cavity. The $53D_{5/2} \rightarrow 54P_{3/2}$ Rydberg transition is driven with the MW electric field at 14.2330 ± 0.0001 GHz. A resonant horn antenna is used to illuminate the Rb vapour cell, perpendicular to the propagation direction of the probe and coupling laser beams (Fig. 1b). The probe and coupling laser beams are circularly polarized with the same polarization, except in Fig. 2 where

they are both linearly polarized parallel to the MW electric field. The MW electric fields are linearly polarized perpendicular to the propagation direction of the laser beams. The change in probe laser transmission is detected using a photodiode, and processed using a lock-in amplifier. The coupling beam intensity is modulated for the lock-in detection. The effective laser linewidths were measured to be 700 ± 200 kHz. The relative jitter of the two lasers is 200 kHz. The data are obtained in three ways. First, the EIT signal at maximum transmission, is monitored as a function of MW electric field strength (Fig. 3). Second, the change in the EIT line shape is recorded as a function of MW frequency for different MW electric field strengths (Fig. 4). The MW electric field frequency is scanned at 20 MHz in 200 ms to acquire these traces. For determining the sensitivity, the probe transmission is monitored on resonance while modulating the MW electric field at a lower frequency. The signal is detected with a second lock-in with a total signal integration time of 1 s. The modulation negligibly improves the signal but is used to measure the sensitivity by comparing the MW on and off signals.

We use a density matrix approach for the four-level ladder system²⁰ (Fig. 1a):

$$\dot{\rho} = -\frac{i}{\hbar}[H, \rho] + L_d \quad (1)$$

where H is the Hamiltonian in the interaction picture. L_d is the dephasing and decay operator. For the calculations, L_d includes spontaneous emission on the $5S_{1/2}(F=2) \rightarrow 5P_{3/2}(F=3)$ transition, $\Gamma_s = 2\pi \times 6.1$ MHz, and transit-time broadening, $\Gamma_{tt} = 2\pi \times 270$ kHz, (ref. 40), because these terms dominate. Spontaneous emission, $\Gamma_R^{54P_{3/2}} \sim 2\pi \times 1$ kHz and $\Gamma_R^{54P_{1/2}} \sim 2\pi \times 0.5$ kHz, black-body decay, $\Gamma_{BB} \sim 2\pi \times 1$ kHz, and collisional dephasing rates, $\Gamma_{coll} \sim 2\pi \times 1$ kHz, of the Rydberg states are small in comparison. The steady-state solutions are velocity averaged using a Maxwell–Boltzmann distribution at room temperature.

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Author contributions

J.P.S. and T.P. conceived the idea. J.P.S. led the project and wrote the paper. J.A.S., A.S. and H.K. carried out the experiments and reduced the data. A.S. and J.A.S. wrote the simulation programs. R.L. contributed useful ideas to the analysis of the experiment. All authors contributed extensively to the work.

Additional information

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Competing financial interests

The authors declare no competing financial interests.